

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA4305

COURSE OUTCOME COVERED

CO -2 : Apply CSS layout and styling techniques to create visually appealing and responsive web interfaces, and use JavaScript to enable interactive client-side behavior. (L3)

Assessment Tool : Design Task

Weightage: 20%

QUESTIONS:

Total Marks: 20

Question 1: Responsive Navigation Bar Design - Marks(4)

Suppose that an e-commerce website requires a responsive navigation bar that adapts across desktop, tablet, and mobile devices.

- (a) Identify key components required in a navigation bar for an e-commerce website.
 - (b) Design a responsive navigation bar using appropriate CSS techniques.
 - (c) Demonstrate how the navigation adapts to smaller screens (e.g., hamburger menu).
 - (d) Evaluate the usability and suggest improvements for better user experience.
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Question 2: Product Hover Effect - Marks(4)

Suppose that an e-commerce website wants to enhance user interaction through visual effects on product items.

- (a) Identify suitable CSS properties for hover effects.
 - (b) Suggest appropriate **CSS animation or transition** for product interaction.
 - (c) Design a hover effect for a product card.
 - (d) Evaluate performance and user experience considerations.
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Question 3: Layout Selection (Flexbox vs Grid) - Marks(4)

Suppose that a dashboard layout needs to be designed for a web application.

- (a) Identify layout requirements of a dashboard (rows, columns, alignment).

- (b) Compare **Flexbox and Grid** based on layout needs.
 - (c) Choose the most suitable layout technique and justify your choice.
 - (d) Evaluate the scalability and responsiveness of the chosen approach.
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Question 4: Mobile-First Blog Layout - Marks(4)

Suppose that a blog page must be designed following a mobile-first approach.

- (a) Identify key sections of a blog layout.
 - (b) Design a mobile-first layout using CSS.
 - (c) Demonstrate how the layout scales for larger screens using media queries.
 - (d) Evaluate accessibility and readability of the design.
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Question 5: JavaScript Event Handling for Dynamic Menu - Marks(4)

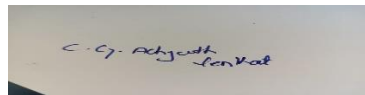
Suppose that a website includes a dynamic menu that responds to user interactions.

- (a) Identify appropriate JavaScript events for menu interaction.
- (b) Design event handling logic for menu toggle functionality.
- (c) Implement interaction for user actions (click/hover).
- (d) Evaluate performance and suggest improvements.

Code of Conduct:

I C G Achyuth Venkat 192311102 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 – Exemplary (4)	Level 3 – Proficient (3)	Level 2 – Developing (2)	Level 1 – Beginning (1)
Understanding of Concept	25	Demonstrates complete understanding of design requirements and concepts	Good understanding with minor gaps	Basic understanding with some errors	Poor understanding or incorrect concepts

Design / Implementation	30	Well-structured, efficient, and responsive design implementation	Correct implementation with minor issues	Basic implementation with inconsistencies	Incorrect or incomplete implementation
Use of Tools (CSS/JS)	20	Effective and advanced use of tools/techniques	Appropriate use of tools	Limited or inconsistent use	Incorrect or minimal use
Analysis & Evaluation	15	Insightful analysis with strong justification and optimization	Good analysis with justification	Basic analysis with limited depth	Poor or no analysis
Documentation / Clarity	10	Clear, well-structured, and properly presented solution	Mostly clear with minor issues	Basic clarity, missing details	Poor or unclear documentation

Question 1: Responsive Navigation Bar Design

(a) Identify key components required in a navigation bar for an e-commerce website.

- Website logo/brand name
- Navigation links (Home, Products, Categories, Contact)
- Search bar
- Shopping cart icon
- User account/login icon
- Hamburger menu for mobile devices

(b) Design a responsive navigation bar using appropriate CSS techniques.

```
nav{
  display:flex;
  justify-content:space-between;
  align-items:center;
  background:#333;
  padding:15px;
}
```

```
.menu{
  display:flex;
  gap:20px;
}
```

```
@media(max-width:768px){
  .menu{
    display:none;
    flex-direction:column;
  }
}
```

```

}
.menu.show{
  display:flex;
}
}

```

(c) Demonstrate how the navigation adapts to smaller screens (e.g., hamburger menu).

```

const menu=document.querySelector(".menu");
const button=document.querySelector(".toggle");

```

```

button.onclick=function(){
  menu.classList.toggle("show");
}

```

(d) Evaluate the usability and suggest improvements for better user experience.

- Easy navigation on all screen sizes.
- Responsive design improves accessibility.
- Add smooth animations for menu transitions.
- Use larger touch targets for mobile users.
- Include keyboard navigation and ARIA labels.

Question 2: Product Hover Effect

(a) Identify suitable CSS properties for hover effects.

- transform
- transition
- box-shadow
- opacity
- background-color

(b) Suggest appropriate CSS animation or transition for product interaction.

```

.card{
  transition:0.3s ease;
}

```

```

.card:hover{
  transform:scale(1.05);
}

```

(c) Design a hover effect for a product card.

```

.card{
  width:250px;
  padding:20px;
  box-shadow:0 0 10px gray;
  transition:0.3s;
}

```

```
.card:hover{
  transform:translateY(-10px);
  box-shadow:0 0 20px black;
}
```

(d) Evaluate performance and user experience considerations.

- Use lightweight animations.
- Avoid excessive motion.
- Improve visual feedback.
- Maintain fast page performance.
- Ensure compatibility across browsers.

Question 3: Layout Selection (Flexbox vs Grid)

(a) Identify layout requirements of a dashboard.

- Header
- Sidebar
- Main content area
- Widgets in rows and columns
- Footer

(b) Compare Flexbox and Grid based on layout needs.

Flexbox

Grid

One-dimensional layout Two-dimensional layout

Best for rows or columns Best for complete page layouts

Easy alignment Precise row and column control

Suitable for navigation Suitable for dashboards

(c) Choose the most suitable layout technique and justify your choice.

```
.container{
  display:grid;
  grid-template-columns:1fr 3fr 1fr;
  gap:20px;
}
```

Justification: CSS Grid is more suitable because dashboards require arranging content in both rows and columns.

(d) Evaluate the scalability and responsiveness of the chosen approach.

- Easily adapts to different screen sizes.
- Supports complex layouts.
- Easy to maintain.
- Works well with media queries.
- Improves responsiveness.

Question 4: Mobile-First Blog Layout

(a) Identify key sections of a blog layout.

- Header
- Navigation

- Blog article
- Sidebar
- Footer

(b) Design a mobile-first layout using CSS.

```
.content{
  display:flex;
  flex-direction:column;
  gap:20px;
}
```

(c) Demonstrate how the layout scales for larger screens using media queries.

```
@media(min-width:768px){
```

```
.content{
  flex-direction:row;
}
```

```
article{
  flex:3;
}
```

```
aside{
  flex:1;
}
```

```
}
```

(d) Evaluate accessibility and readability of the design.

- Responsive on all devices.
- Easy to read with proper spacing.
- Uses semantic HTML elements.
- Supports keyboard navigation.
- Provides better accessibility.

Question 5: JavaScript Event Handling for Dynamic Menu

(a) Identify appropriate JavaScript events for menu interaction.

- click
- mouseover
- mouseout
- mouseenter
- mouseleave

(b) Design event handling logic for menu toggle functionality.

```
button.addEventListener("click",function(){
  menu.classList.toggle("show");
});
```

(c) Implement interaction for user actions (click/hover).

```
menuItem.addEventListener("mouseover",function(){  
    this.style.background="green";  
});
```

```
menuItem.addEventListener("mouseout",function(){  
    this.style.background="blue";  
});
```

(d) Evaluate performance and suggest improvements.

- Use event delegation for multiple menu items.
- Minimize DOM manipulation.
- Improve accessibility with ARIA attributes.
- Optimize JavaScript execution.
- Test across different browsers and devices.